Problem sheet 7 for Rigid Analytic Geometry Summer Term 2026

We retain the notations and assumptions from Problem 6 of the previous sheet. Thus, K is assumed to be algebraically closed and complete with respect to the ultrametric non-discrete absolute value $|\cdot|$. It follows from these assumptions that $|K^\times|$ is dense in $[0,\infty)$.

For $x \in K$ and $R \in (0,\infty)_R$, let

$$B_{\leq R}(x) = \big\{ y \in K^\circ \;\big|\; |x-y| \leq R \big\}$$
$$B_{<R}(x) = \big\{ y \in K^\circ \;\big|\; |x-y| < R \big\}$$

Both are open subsets of K and it follows from Fact 3.1.1 of the lecture that they are in fact both clopen. If $R \notin |K^\times|$ they conincide, otherwise the second is a proper subset of the first.

Problem 1 (4 points)**.** *In the situation of Problem 6 from sheet 6, show that there is a unique van der Put point ξ of $X = \mathrm{Sp}\mathbb{T}_1$ such that a rational open subset $\Omega = \mathcal{R}(f_0|f_1,\ldots,f_m)$ belongs to ξ if and only if the first point of the alternative from that Problem occurs:*

- *We have $N_R(f_j) \leq N_R(f_0)$ for $1 \leq j \leq m$ or, equivalently, there are $r_- \in (0,R)_\mathbb{R}$ and $r_+ \in (R,1)_\mathbb{R}$ such that $B_{\leq r_+}(0) \subseteq \Omega \cup B_{\leq r_-}(0)$.*
- *There is a positive integer $j \leq m$ such that $N_R(f_j) > N_R(f_0)$, or, equivalently, there are $r_- \in (0,R)_\mathbb{R}$ and $r_+ \in (R,1)_\mathbb{R}$ such that $\Omega \cap B_{\leq r+}(0) \subseteq B_{\leq r_-}(0)$.*

Remark 1. *In each case of the alternative, the second condition (which is equivalent to the first one) depends only on Ω. Thus, the set $\xi_{\mathfrak{Rat}_{\mathbb{T}_1}}$ of rational open subsets satisfying the first alternative is well-defined, and it suffices to show that Problems 2 and 3 from the previous can be applied.*

Remark 2. *Of course, to this result one can apply translation by any $x \in K^\circ$. Thus, for every $R \in (0,1)_\mathbb{R} \setminus K^\times$, there is a van der Put point ξ containing precisely the rational open subsets Ω for which there are $r_- \in (0,R)_\mathbb{R}$ and $r_+ \in (R,1)_\mathbb{R}$ such that $B_{\leq r_+}(x) \subseteq \Omega \cup B_{\leq r_-}(x)$. Moreover, for every other rational open subset $\Omega \cap B_{\leq r+}(x) \subseteq B_{\leq r_-}(x)$. This van der Put point corresponds to the valuation*

$$\mathbb{T}_1 \xrightarrow{f \to N_R\big(f(\cdot - x)\big)} [0,\infty)_\mathbb{R}$$

in the sense that $\mathcal{R}(f_0|f_1,\ldots,f_m) \in \xi$ if and only if $N_R(f_j) \leq N_R(f_0)$ for $1 \leq j \leq m$. Note that x is not uniquely determined by ξ, but R and the ball $B = B_{\leq R}(x) = B_{<r}(x)$ are. We will denote ξ by ξ_B.

Problem 2 (2 points)**.** *If $x \in K^\circ$ and $R \in (0,1]_{\mathbb{R}} \cap K^\times$, show that $B_{\leq R}(x) \in \mathfrak{Rat}_{\mathbb{T}_1}$.*

Problem 3 (4 points)**.** *In the situation of the first problem, let v be a van der Put point of X such that for $\rho \in (0,1]_{\mathbb{R}} \cap |K^\times|$, v contains $K_\rho(0)$ if and only if $\rho > R$. Show that $v = \xi$.*

Remark 3. *Analogously, in the situation of the previous remark where we considered arbitary balls B of radius $R \in (0,1)_{\mathbb{R}} \setminus |K^\times|$, ξ_B is the unique van der Put point containing all $\leq$-balls of radius $\rho \in (0,1]_{\mathbb{R}} \cap K^\times$ containing B but no such ball which is contained in B.*

This suggests to attempt to find all van der Put point ξ of X by investigating which balls of radius $\rho \in (0,1]_{\mathbb{R}} \cap K^\times$ are elements of ξ. For each ξ we will find a valuation v of $\mathbb{T}_1$ such that

(1) $\mathcal{R}(f_0|f_1,\ldots,f_m) \in \xi$ if and only if $v(f_j) \leq v(f_0)$ for $1 \leq j \leq m$,

and the equivalence class of v belongs to $\mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$. The resulting description of X^* (which can be seen to be in bijection with $\mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$) is the one given in P. Scholze, Perfectoid Spaces, Example 2.20. It can also be compared with the beginning of subsection 7.2 of the 2003 english edition of the textbook by Fresnel and van der Put. However, Fresnel and van der Put only describe the points relevant to the Berkovich picture and their approach is less geometric as they consider the valuations directly rather than studying geometric properties of the van der Put point.

It is necessary to start with a few general considerations about balls in K°. For $R \in (0,1]_{\mathbb{R}}$, let

$$\mathfrak{K}_{\leq R} = \left\{ K_{\leq R}(x) \;\middle|\; x \in K^\circ \right\}$$
$$\mathfrak{K}_{< R} = \left\{ K_{< R}(x) \;\middle|\; x \in K^\circ \right\}.$$

Furthermore, let

$$\mathfrak{K}_{\leq} = \bigcup_{R \in (0,1]_{\mathbb{R}}} \mathfrak{K}_{\leq R}$$
$$\mathfrak{K}_{<} = \bigcup_{R \in (0,1]_{\mathbb{R}}} \mathfrak{K}_{< R}$$

and $\mathfrak{K} = \mathfrak{K}_{\leq} \cup \mathfrak{K}_{<}$.

Problem 4 (4 points)**.** *For arbitrary $B_{1,2} \in \mathfrak{K}$, precisely one of the four cases $B_1 \cap B_2 = \emptyset$, $B_1 \subset B_2$, $B_2 \subset B_1$ or $B_1 = B_2$ occurs.*

We will continue this program of describing X^* on the next sheets. The remainder of the current sheet is devoted to a few omitted proofs

from the lecture. It is suggested to work them out in the order presented here, always trying to use a previously given proof where this seems to make sense. The first 3 points from Fact 3.1.3 of the lecture can be used.

Let always A be a topological ring and M a topological A-module, and let K be any field which is complete and non-discrete with respect to an ultrametric absolute value $|\cdot|$.

Problem 5 (1 point)**.** *If $X \subseteq A$ and $Y \subseteq M$ are bounded, XY is a bounded subset of M.*

Problem 6 (1 point)**.** *If $X \subseteq A$ is bounded then all X^n with $n \in \mathbb{N}$ are bounded.*

Problem 7 (1 point)**.** *Every $X \subseteq A$ which is bounded and topologically nilpotent is power-bounded.*

Problem 8 (2 points)**.** *Let $R = K \oplus K \cdot \varepsilon$ with $\varepsilon^2 = 0$, equipped with the topology of K^2. Show that $X = K \cdot \varepsilon$ is a subset of R which is topologically nilpotent but not power-bounded.*

Problem 9 (1 point)**.** *Let $X, Y \subseteq A$ be topologically nilpotent. Show that $X \cup Y$ is topologically nilpotent!*

Problem 10 (1 point)**.** *Let $X, Y \subseteq A$ be topologically nilpotent. Show that XY is topologically nilpotent!*

Problem 11 (2 points)**.** *Let $X \subseteq A$ be power-bounded and $Y \subseteq A$ be topologically nilpotent. Show that XY is topologically nilpotent!*

Of the 23 points from this sheet, three are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday June 15. Good luck!